

6th Meeting 12/2

The dataset that i gather for normal yolo detection

Detection

```
val: New cache created: /media/newadmin/master/popw-protocol/workernet/refined workernet/pose_proper_split/test/labels.cache
=====  
Class      Images  Instances  Box(P   R   mAP50  mAP50-95): 100%| 14/14 [00:01<00:00, 11.49it/s]  
all        109      293      0.88    0.927  0.936   0.617  
person     89       91      0.979   0.978  0.975   0.828  
bottle     89       97      0.848   0.979  0.984   0.675  
barcode    20       20      0.922   1      0.995   0.776  
cap_closed 50       52      0.806   0.799  0.812   0.373  
cap_open   33       33      0.844   0.879  0.914   0.431  
Speed: 0.9ms preprocess, 2.9ms inference, 0.0ms loss, 0.9ms postprocess per image  
Results saved to detection_proper_split/run3  
  
===== RESULTS =====  
  
Dataset:  
Train: 317 images  
Val: 77 images  
Test: 110 images  
  
Validation:  
mAP@50: 0.961  
mAP@50-95: 0.635
```

Pose multiclass head (3 class)

```
===== TEST =====  
Ultralytics YOLOv8 2.103 Python-3.10.19 torch-2.0.1+cu128 CUDA:0 (NVIDIA GeForce RTX 3060, 12044MiB)  
val: Scanning /media/newadmin/master/popw-protocol/workernet/refined workernet/pose_proper_split/test/labels... 94 images, 0 backgrounds, 0 corrupt: 100%| 94/94 [00:  
val: New cache created: /media/newadmin/master/popw-protocol/workernet/refined workernet/pose_proper_split/test/labels.cache  
=====  
Class      Images  Instances  Box(P   R   mAP50  mAP50-95): 100%| 12/12 [00:01<00:00, 8.87  
all        94       94      0.984   0.992  0.99   0.919  
checking   41       41      0.973   1      0.98   0.925  
rotation   42       42      0.998   0.976  0.995  0.936  
storing    11       11      0.982   1      0.995  0.896  
Speed: 0.5ms preprocess, 4.3ms inference, 0.0ms loss, 1.2ms postprocess per image  
Results saved to pose_proper_split/run3  
  
===== RESULTS =====  
  
Dataset:  
Train: 202 images  
Val: 20 images  
Test: 94 images  
  
Validation:  
Activity Box mAP@50: 0.995  
Activity Box mAP@50-95: 0.937  
Pose mAP@50: 0.995  
Pose mAP@50-95: 0.929  
  
Test:  
Activity Box mAP@50: 0.990  
Activity Box mAP@50-95: 0.934  
Pose mAP@50: 0.990  
Pose mAP@50-95: 0.919
```

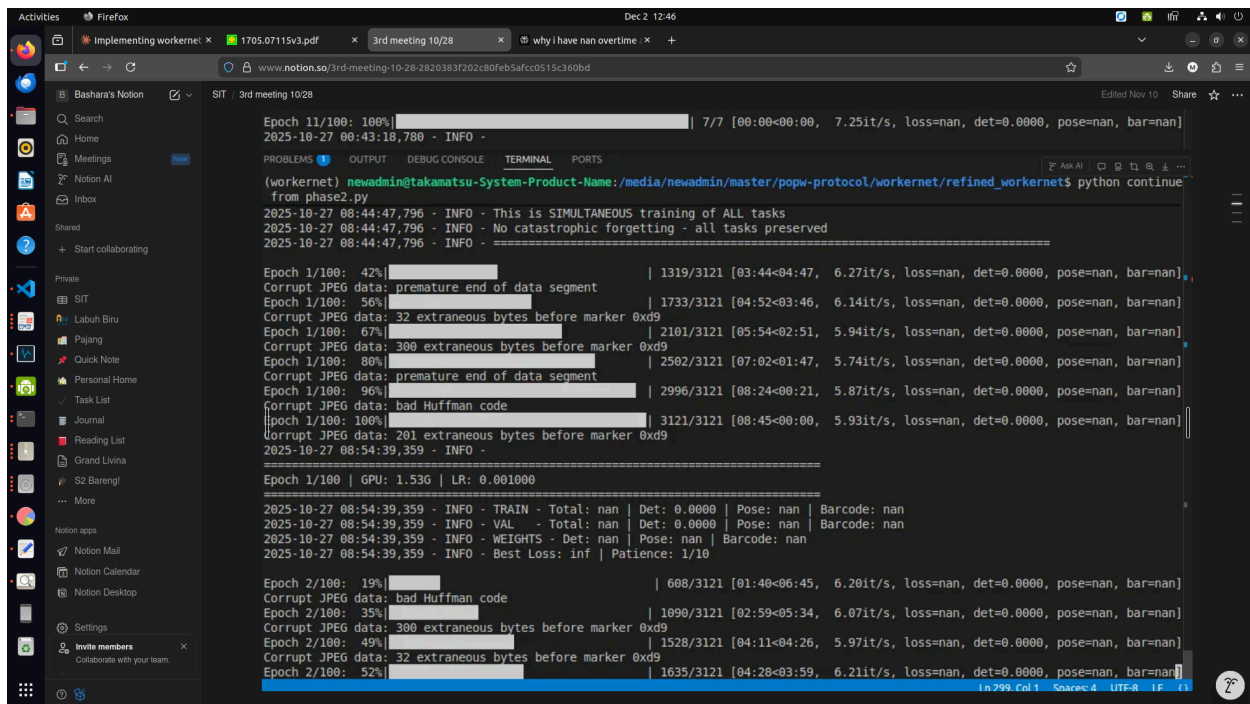
So, I try to implement in

arxiv.org

<https://arxiv.org/pdf/1705.07115>

<https://arxiv.org/pdf/1705.07115>

The problem from last month finally found



```
Epoch 11/100: 100% | 7/7 [00:00<00:00, 7.25it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
2025-10-27 00:43:18,780 - INFO -
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(workernet) newadmin@takamatsu-System-Product-Name:/media/newadmin/master/popw-protocol/workernet/refined_workernet$ python continue
from phase2.py
2025-10-27 08:44:47,796 - INFO - This is SIMULTANEOUS training of ALL tasks
2025-10-27 08:44:47,796 - INFO - No catastrophic forgetting - all tasks preserved
2025-10-27 08:44:47,796 - INFO -
Epoch 1/100: 42% | 1319/3121 [03:44<04:47, 6.27it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: premature end of data segment | 1733/3121 [04:52<03:46, 6.14it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Epoch 1/100: 56% | 2101/3121 [05:54<02:51, 5.94it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: 32 extraneous bytes before marker 0xd9 | 2502/3121 [07:02<01:47, 5.74it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Epoch 1/100: 67% | 2996/3121 [08:24<00:21, 5.87it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: premature end of data segment | 3121/3121 [08:45<00:00, 5.93it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Epoch 1/100: 80% | 2502/3121 [07:02<01:47, 5.74it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: 300 extraneous bytes before marker 0xd9
Epoch 1/100: 96% | 2996/3121 [08:24<00:21, 5.87it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: bad Huffman code
Epoch 1/100: 100% | 3121/3121 [08:45<00:00, 5.93it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: 201 extraneous bytes before marker 0xd9
2025-10-27 08:54:39,359 - INFO -
Epoch 1/100 | GPU: 1.53G | LR: 0.001000
2025-10-27 08:54:39,359 - INFO - TRAIN - Total: nan | Det: 0.0000 | Pose: nan | Barcode: nan
2025-10-27 08:54:39,359 - INFO - VAL - Total: nan | Det: 0.0000 | Pose: nan | Barcode: nan
2025-10-27 08:54:39,359 - INFO - WEIGHTS - Det: nan | Pose: nan | Barcode: nan
2025-10-27 08:54:39,359 - INFO - Best Loss: inf | Patience: 1/10
Epoch 2/100: 19% | 608/3121 [01:40<06:45, 6.20it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: bad Huffman code | 1090/3121 [02:59<05:34, 6.07it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Epoch 2/100: 35% | 1090/3121 [02:59<05:34, 6.07it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: 300 extraneous bytes before marker 0xd9 | 1528/3121 [04:11<04:26, 5.97it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Epoch 2/100: 49% | 1528/3121 [04:11<04:26, 5.97it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Corrupt JPEG data: 32 extraneous bytes before marker 0xd9 | 1635/3121 [04:28<03:59, 6.21it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
Epoch 2/100: 52% | 1635/3121 [04:28<03:59, 6.21it/s, loss=nan, det=0.0000, pose=nan, bar=nan]
```

```
Total train samples: 49927
--- Epoch 1/100 ---
Epoch 1 Training: 8% | 239/3120 [00:14<02:43, 17.64it/s, loss=36.3, det=43.5, barcode=2.29, pose=3.64e-6]
Corrupt JPEG data: premature end of data segment | 499/3120 [00:29<02:28, 17.67it/s, loss=28.3, det=43.9, barcode=2.3, pose=1.19e-6]
Epoch 1 Training: 16% | 499/3120 [00:29<02:28, 17.67it/s, loss=28.3, det=43.9, barcode=2.3, pose=1.19e-6]
Corrupt JPEG data: 32 extraneous bytes before marker 0xd9 | 1419/3120 [01:23<01:39, 17.09it/s, loss=12, det=44.6, barcode=2.3, pose=1.51e-7]
Epoch 1 Training: 45% | 1419/3120 [01:23<01:39, 17.09it/s, loss=12, det=44.6, barcode=2.3, pose=1.51e-7]
Corrupt JPEG data: premature end of data segment | 1551/3120 [01:30<01:31, 17.13it/s, loss=8.23, det=33, barcode=2.3, pose=1.19e-8]
Epoch 1 Training: 50% | 1551/3120 [01:30<01:31, 17.13it/s, loss=8.23, det=33, barcode=2.3, pose=1.19e-8]
Corrupt JPEG data: 300 extraneous bytes before marker 0xd9 | 2663/3120 [02:35<00:27, 16.88it/s, loss=5.48, det=46, barcode=2.31, pose=8.37e-10]
Epoch 1 Training: 85% | 2663/3120 [02:35<00:27, 16.88it/s, loss=5.48, det=46, barcode=2.31, pose=8.37e-10]
Corrupt JPEG data: bad Huffman code | 3120/3120 [03:01<00:00, 17.17it/s, loss=5.06, det=54.8, barcode=2.31, pose=1.24e-9]
Epoch 1 Training: 100% | 3120/3120 [03:01<00:00, 17.17it/s, loss=5.06, det=54.8, barcode=2.31, pose=1.24e-9]
Epoch 1 avg_train_loss: 14.653672
--- Generating Comparison Report ---
Time Comparison:
Traditional: 19950.12s
WorkerNet: 0.00s
Time Saved: 19950.12s (100.00%)
Performance Comparison (mAP50):
Traditional - barcode: 0.9777
Traditional - bottle: 0.8879
Traditional - person: 0.6252
Traditional - pose: 0.0000
WorkerNet (Unified): 0.0000
```

The importance of the gradient adjustment loss is to make sure the

Small Loss (0.001)



Large Gradient (because $\text{gradient} = 1 \div \text{loss}$)



Huge Parameter Update (update (backbone, neck, head [detection pose activity])
= $\text{learning_rate} \times \text{gradient}$)



Parameters pushed to wrong place



Model tries impossible math (log of negative, sqrt of negative, $1 \div 0$)



NaN appears! CRASH!

Detection: Loss = 1.8460 → Gradient ≈ 0.5 (large)

Pose: Loss = 0.2509 → Gradient ≈ 0.05 (10x smaller!) → epoch 100: 0.0005 →
gradient is higher → update (backbone etc) high → parameters changed → makes
the loss function high

Activity: Loss = 0.7619 → Gradient ≈ 0.3 (medium)

Adjust each class gradient to prevent the update to be stalling

stabilize the gradient (it can see the current situation) gradient would not be really
high

At the first, I try to train from the detection and pose data that I gather

detection → 5 classes (person, bottle, barcode, cap close, cap open)

pose → activity classes (checking, rotating, storing) + 13 keypoints. But it was not
the main goal of workernet (it supposedly to be efficient in one train)

```

2025-12-02 12:51:58,135 | INFO | =====
2025-12-02 12:51:58,136 | INFO | WORKERNET UNIFIED MODEL - TRAINING (FIXED)
2025-12-02 12:51:58,136 | INFO | =====
Data directories:
2025-12-02 12:51:58,136 | INFO | Detection: detection_proper_split
2025-12-02 12:51:58,136 | INFO | Pose: pose_proper_split
Loading datasets...
2025-12-02 12:51:58,140 | INFO | Loaded 317 detection samples from detection_proper_split/train
2025-12-02 12:51:58,141 | INFO | Loaded 77 detection samples from detection_proper_split/val
2025-12-02 12:51:58,144 | INFO | Loaded 202 pose samples from pose_proper_split/train
2025-12-02 12:51:58,145 | INFO | Loaded 28 pose samples from pose_proper_split/val
2025-12-02 12:51:58,145 | INFO |
Dataset sizes:
2025-12-02 12:51:58,145 | INFO | Detection train: 317
2025-12-02 12:51:58,145 | INFO | Detection val: 77
2025-12-02 12:51:58,145 | INFO | Pose train: 202
2025-12-02 12:51:58,145 | INFO | Pose val: 28
2025-12-02 12:51:58,145 | INFO |
Combined dataset:
2025-12-02 12:51:58,145 | INFO | Train: 519
2025-12-02 12:51:58,145 | INFO | Val: 105

```

```

(workernet clean) newadmin@takamatsu-System-Product-Name:/media/newadmin/master/popw-protocol/workernet/refined_workernet$ python train_workernet_fixed.py
Epoch 1: 100% | 64/64 [00:19<00:00, 3.28it/s, act=0.5851, det=1.5217, pose=0.1656, total=1.0884]
Validating: 100% | 14/14 [00:01<00:00, 8.62it/s, act=5.8538, det=0.0000, pose=0.1282, total=3.0746]
2025-12-02 12:52:22,177 | INFO | Epoch 1/100 | Train Total: 1.4837 | Val Total: 1.2784
2025-12-02 12:52:22,177 | INFO | Train - Det: 1.9381 | Pose: 0.3277 | Act: 0.7492
2025-12-02 12:52:22,177 | INFO | Val - Det: 1.0684 | Pose: 0.0698 | Act: 1.4336
2025-12-02 12:52:22,178 | INFO | LR: 0.001000 | Weights Det:0.948 Pose:1.067 Act:1.046
2025-12-02 12:52:22,561 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:52:22,562 | INFO | New best model saved! Loss: 1.2784
Epoch 2: 100% | 64/64 [00:17<00:00, 3.58it/s, act=0.5179, det=1.3489, pose=0.1404, total=0.9131]
Validating: 100% | 14/14 [00:01<00:00, 9.92it/s, act=1.3165, det=0.0000, pose=0.0868, total=0.6585]
2025-12-02 12:52:41,842 | INFO | Epoch 2/100 | Train Total: 1.0367 | Val Total: 0.6430
2025-12-02 12:52:41,842 | INFO | Train - Det: 1.3947 | Pose: 0.1279 | Act: 0.6781
2025-12-02 12:52:41,842 | INFO | Val - Det: 0.8008 | Pose: 0.0409 | Act: 0.5330
2025-12-02 12:52:41,842 | INFO | LR: 0.000999 | Weights Det:0.921 Pose:1.145 Act:1.100
2025-12-02 12:52:42,272 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:52:42,272 | INFO | New best model saved! Loss: 0.6430
Epoch 3: 100% | 64/64 [00:17<00:00, 3.59it/s, act=0.4682, det=1.1847, pose=0.1207, total=0.7569]
Validating: 100% | 14/14 [00:01<00:00, 10.09it/s, act=3.0140, det=0.0000, pose=0.0944, total=1.6262]
2025-12-02 12:53:01,512 | INFO | Epoch 3/100 | Train Total: 0.8988 | Val Total: 0.6534
2025-12-02 12:53:01,512 | INFO | Train - Det: 1.2620 | Pose: 0.1246 | Act: 0.6173
2025-12-02 12:53:01,512 | INFO | Val - Det: 0.6717 | Pose: 0.0420 | Act: 0.6867
2025-12-02 12:53:01,512 | INFO | LR: 0.000998 | Weights Det:0.904 Pose:1.226 Act:1.156
Epoch 4: 100% | 64/64 [00:17<00:00, 3.57it/s, act=0.4209, det=1.1690, pose=0.0928, total=0.6616]
Validating: 100% | 14/14 [00:01<00:00, 9.85it/s, act=0.8228, det=0.0000, pose=0.0871, total=0.3247]
2025-12-02 12:53:20,860 | INFO | Epoch 4/100 | Train Total: 0.8000 | Val Total: 0.4757
2025-12-02 12:53:20,860 | INFO | Train - Det: 1.2058 | Pose: 0.1103 | Act: 0.5653
2025-12-02 12:53:20,860 | INFO | Val - Det: 0.6959 | Pose: 0.0395 | Act: 0.4053
2025-12-02 12:53:20,860 | INFO | LR: 0.000996 | Weights Det:0.893 Pose:1.309 Act:1.224
2025-12-02 12:53:21,278 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:53:21,278 | INFO | New best model saved! Loss: 0.4757
Epoch 5: 100% | 64/64 [00:17<00:00, 3.58it/s, act=0.6615, det=1.2297, pose=0.1102, total=0.8157]
Validating: 100% | 14/14 [00:01<00:00, 9.95it/s, act=0.8382, det=0.0000, pose=0.0835, total=0.3040]
2025-12-02 12:53:40,576 | INFO | Epoch 5/100 | Train Total: 0.7787 | Val Total: 0.4531
2025-12-02 12:53:40,577 | INFO | Train - Det: 1.1875 | Pose: 0.1189 | Act: 0.5985
2025-12-02 12:53:40,577 | INFO | Val - Det: 0.6474 | Pose: 0.0383 | Act: 0.4358
2025-12-02 12:53:40,577 | INFO | LR: 0.000994 | Weights Det:0.886 Pose:1.397 Act:1.277
2025-12-02 12:53:41,022 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:53:41,023 | INFO | New best model saved! Loss: 0.4531
Epoch 6: 100% | 64/64 [00:18<00:00, 3.55it/s, act=0.4619, det=1.2145, pose=0.0951, total=0.6347]
Validating: 100% | 14/14 [00:01<00:00, 9.11it/s, act=1.9888, det=0.0000, pose=0.0946, total=1.0571]
2025-12-02 12:54:00,615 | INFO | Epoch 6/100 | Train Total: 0.7011 | Val Total: 0.5568
2025-12-02 12:54:00,615 | INFO | Train - Det: 1.1446 | Pose: 0.1086 | Act: 0.5558
2025-12-02 12:54:00,615 | INFO | Val - Det: 0.6806 | Pose: 0.0420 | Act: 0.5936
2025-12-02 12:54:00,615 | INFO | LR: 0.000991 | Weights Det:0.883 Pose:1.487 Act:1.338
Epoch 7: 100% | 64/64 [00:18<00:00, 3.47it/s, act=0.3572, det=1.2428, pose=0.1171, total=0.5571]
Validating: 100% | 14/14 [00:01<00:00, 9.28it/s, act=2.5070, det=0.0000, pose=0.1450, total=1.4721]
2025-12-02 12:54:20,578 | INFO | Epoch 7/100 | Train Total: 0.6583 | Val Total: 0.7370
2025-12-02 12:54:20,578 | INFO | Train - Det: 1.1464 | Pose: 0.1103 | Act: 0.5381
2025-12-02 12:54:20,578 | INFO | Val - Det: 0.6549 | Pose: 0.0424 | Act: 0.8854
2025-12-02 12:54:20,578 | INFO | LR: 0.000988 | Weights Det:0.882 Pose:1.580 Act:1.399

```

NVIDIA-SMI 535.274.02			Driver Version: 535.274.02			CUDA Version: 12.2		
GPU	Name	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr. ECC		
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage	GPU-Util	Compute M.		
						MIG M.		
0	NVIDIA GeForce RTX 3060	Off	00000000:02:00:0	Off			N/A	
30%	59C	P2	74W / 170W	2093MiB / 12288MiB	42%	Default	N/A	

Processes:							
GPU	GI	CI	PID	Type	Process name	GPU Memory	
	ID	ID				Usage	
0	N/A	N/A	5185	G	/usr/lib/xorg/Xorg	4MiB	
0	N/A	N/A	3594061	C	python	2080MiB	

The training has no problem, but i want to use the advantage of this model. Learning everything simultaneously. So the data becoming slimmer (divide it by 2) 1 image has 2 labels for detection and pose. Also for the classes activity. Training becoming faster. the load becoming more efficient.

```

Loading datasets...
2025-12-02 12:54:38,962 | INFO | Loaded 204 samples from train set
2025-12-02 12:54:38,963 | INFO | Loaded 43 samples from val set
2025-12-02 12:54:38,963 | INFO | Train: 204 samples
2025-12-02 12:54:38,963 | INFO | Val: 43 samples
Creating WorkerNet model...

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```

Epoch 1: 100% | 26/26 | 00:10:00:00, 2.45it/s, act=0.9147, det=1.6789, pose=0.1524, total=1.3551]
Validating: 100% | 6/6 | 00:01:00:00, 5.65it/s, act=0.8179, det=4.1905, pose=1.5133, total=3.2316]
2025-12-02 12:57:38,533 | INFO | Epoch 1/100 | Train: 1.7039 | Val: 4.1944 | Det: 2.0941 | Pose: 0.3426 | Act: 0.9900 | LR: 0.001800 | Weights Det:0.977 Pose:1.027 Act:0.992
2025-12-02 12:57:38,959 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:57:38,960 | INFO | New best model! Loss: 4.1944
Epoch 2: 100% | 26/26 | 00:09:00:00, 2.82it/s, act=0.2843, det=1.4536, pose=0.1235, total=0.8905]
Validating: 100% | 6/6 | 00:01:00:00, 5.65it/s, act=0.3579, det=1.4849, pose=0.1165, total=0.9385]
2025-12-02 12:57:49,245 | INFO | Epoch 2/100 | Train: 1.1135 | Val: 1.3797 | Det: 1.5175 | Pose: 0.1458 | Act: 0.6170 | LR: 0.000999 | Weights Det:0.960 Pose:1.057 Act:1.024
2025-12-02 12:57:49,666 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:57:49,666 | INFO | New best model! Loss: 1.3797
Epoch 3: 100% | 26/26 | 00:08:00:00, 2.90it/s, act=0.6616, det=1.3364, pose=0.1175, total=1.0045]
Validating: 100% | 6/6 | 00:00:00:00, 6.46it/s, act=0.0025, det=1.5952, pose=0.3762, total=0.9178]
2025-12-02 12:57:59,569 | INFO | Epoch 3/100 | Train: 0.9241 | Val: 1.9175 | Det: 1.3676 | Pose: 0.1234 | Act: 0.4550 | LR: 0.000998 | Weights Det:0.948 Pose:1.087 Act:1.060
Epoch 4: 100% | 26/26 | 00:09:00:00, 2.71it/s, act=0.2267, det=1.2402, pose=0.0998, total=0.6937]
Validating: 100% | 6/6 | 00:00:00:00, 6.07it/s, act=22.8614, det=1.4493, pose=0.1086, total=13.1860]
2025-12-02 12:58:10,150 | INFO | Epoch 4/100 | Train: 0.8265 | Val: 5.6160 | Det: 1.2937 | Pose: 0.1175 | Act: 0.3858 | LR: 0.000996 | Weights Det:0.938 Pose:1.117 Act:1.095
Epoch 5: 100% | 26/26 | 00:09:00:00, 2.72it/s, act=1.4890, det=1.2378, pose=0.1097, total=1.3805]
Validating: 100% | 6/6 | 00:01:00:00, 5.71it/s, act=0.3370, det=1.4292, pose=0.1080, total=0.8235]
2025-12-02 12:58:20,781 | INFO | Epoch 5/100 | Train: 0.7942 | Val: 1.4736 | Det: 1.2489 | Pose: 0.1115 | Act: 0.4104 | LR: 0.000994 | Weights Det:0.930 Pose:1.148 Act:1.128
Epoch 6: 100% | 26/26 | 00:09:00:00, 2.75it/s, act=1.5862, det=1.2523, pose=0.1391, total=1.4634]
Validating: 100% | 6/6 | 00:01:00:00, 5.89it/s, act=1.2911, det=1.2118, pose=0.1009, total=1.2510]
2025-12-02 12:58:31,254 | INFO | Epoch 6/100 | Train: 0.7219 | Val: 0.9054 | Det: 1.2070 | Pose: 0.1158 | Act: 0.3485 | LR: 0.000991 | Weights Det:0.924 Pose:1.179 Act:1.160
2025-12-02 12:58:31,681 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:58:31,682 | INFO | New best model! Loss: 0.9054
Epoch 7: 100% | 26/26 | 00:10:00:00, 2.60it/s, act=0.4669, det=1.1868, pose=0.1335, total=0.7637]
Validating: 100% | 6/6 | 00:01:00:00, 5.85it/s, act=0.0719, det=1.2493, pose=0.0940, total=0.5324]
2025-12-02 12:58:42,715 | INFO | Epoch 7/100 | Train: 0.6532 | Val: 0.7334 | Det: 1.1774 | Pose: 0.1092 | Act: 0.2963 | LR: 0.000988 | Weights Det:0.919 Pose:1.211 Act:1.192
2025-12-02 12:58:43,144 | INFO | Checkpoint saved: runs/workernet_unified/best_workernet.pt
2025-12-02 12:58:43,144 | INFO | New best model! Loss: 0.7334
Epoch 8: 100% | 26/26 | 00:09:00:00, 2.78it/s, act=0.0732, det=1.1268, pose=0.1034, total=0.4586]
Validating: 100% | 6/6 | 00:00:00:00, 6.02it/s, act=0.9359, det=1.2129, pose=0.0917, total=1.0193]
2025-12-02 12:58:53,500 | INFO | Epoch 8/100 | Train: 0.5766 | Val: 1.0776 | Det: 1.1710 | Pose: 0.1081 | Act: 0.2103 | LR: 0.000984 | Weights Det:0.914 Pose:1.243 Act:1.228
Epoch 9: 100% | 26/26 | 00:09:00:00, 2.78it/s, act=0.1575, det=1.1602, pose=0.0971, total=0.4992]
Validating: 100% | 6/6 | 00:01:00:00, 5.60it/s, act=7.4031, det=1.2246, pose=0.0914, total=5.0941]
2025-12-02 12:59:03,948 | INFO | Epoch 9/100 | Train: 0.5832 | Val: 1.9231 | Det: 1.1212 | Pose: 0.1062 | Act: 0.2953 | LR: 0.000980 | Weights Det:0.912 Pose:1.276 Act:1.262
Epoch 10: 100% | 26/26 | 00:09:00:00, 2.73it/s, act=0.5448, det=1.1219, pose=0.0943, total=0.7081]
Validating: 100% | 6/6 | 00:01:00:00, 5.79it/s, act=1.1393, det=1.2677, pose=0.0901, total=1.1571]

```

NVIDIA-SMI 535.274.02			Driver Version: 535.274.02			CUDA Version: 12.2		
GPU	Name		Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr. ECC	
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage		GPU-Util	Compute M.	
							MIG M.	
0	NVIDIA GeForce RTX 3060		Off	00000000:02:00.0	Off		N/A	
30%	57C	P2	62W / 170W	2101MiB / 12288MiB		38%	Default	N/A
							N/A	
Processes:								
GPU	GI	CI	PID	Type	Process name	GPU Memory		
	ID	ID				Usage		
0	N/A	N/A	5185	G	/usr/lib/xorg/Xorg	4MiB		
0	N/A	N/A	3591852	C	python	2088MiB		

Short Term Planning (4)

Aa Name	📅 Date	☰ Tags
<u>Untitled</u>		
<u>Write paper and submission</u>	@December 31, 2025 → January 31, 2026	
<u>Address feedback, revise code and documentation</u>		
<u>Prepare figures, diagrams, and tables for paper</u>		
<u>POPW Implementation COMPLETE</u>	@December 22, 2025 → December 30, 2025	
<u>Data and result collection for paper</u>	@December 8, 2025 → December 14, 2025	
<u>Address feedback, revise code and documentation</u>		
<u>Prepare figures, diagrams, and tables for paper</u>		
<u>Address feedback, revise code and documentation</u>		
<u>Prepare figures, diagrams, and tables for paper</u>		
<u>Optimization</u>	@December 2, 2025 → December 14, 2025	

Aa Name	📅 Date	☰ Tags
<u>Real world lab testing</u>	@December 2, 2025 → December 21, 2025	
<u>POPW protocol implementation</u>	@December 2, 2025 → December 21, 2025	
<u>Implement the spatiotemporal fingerprint</u>	@December 2, 2025 → December 21, 2025	
<u>Debugging</u>	@November 11, 2025 → December 14, 2025	
<u>Untitled</u>		
<u>Data Collection</u>	@November 11, 2025 → December 8, 2025	
<u>Finalize architecture and prepare result</u>	@October 31, 2025 → December 8, 2025	
<u>Complete workernet training & benchmarking</u>	@October 14, 2025 → December 8, 2025	
<u>Untitled</u>	@November 17, 2023 → November 30, 2023	

Explain gradient 1 / loss function value.

Explain loss function that i use for this model

How did they adjust the value of this gradient (in the paper)

If it has any improvement → model

I will breakdown the model archictural workernet (what is the improvement of this mdoel → compare to the base model)

Write the mathematical equation function loss (explain the normal loss function)

I will research about the conference that matched my project → look wiki cfp call for paper (for asia only)